

User Guide: From CAD to a Simulation-Ready SDF Model

Section 1: Introduction and SDF Overview

This guide explains how to convert a 3D model from CAD software (e.g., SolidWorks, Fusion 360, Inventor) into an SDF (Simulation Description Format) file ready for import into Flowstate. This is necessary because Flowstate currently supports importing robots or equipment with movable components via SDF files, but not their creation within the tool itself.

What is an SDF file? SDF is an XML file format that describes all aspects of a simulation, including robots, lights, sensors, and the environment. Think of it as a blueprint for a virtual world. It's a core component of the Gazebo robotics simulator, which is Flowstate's primary simulator.

Here's how you'd typically author an SDF file:

- **Use a Plain Text Editor:** As SDF files are XML, any text editor (e.g., VS Code, Sublime Text, Notepad) can be used. Plugins for many editors offer XML syntax highlighting.
- **Start with SDF Tags:** Every SDF file begins and ends with `<sdf>` tags, where you specify the SDF format version.
- **Define a Model:** Inside the `<sdf>` tags, define a `<model>` to represent your robot, furniture, or any simulation object.
- **Add Links and Joints:** A robot model consists of links (rigid parts) and joints (connecting links and enabling movement). Define each of these.
- **Define Visual and Collision Properties:** For each link, specify a `<visual>` tag (appearance) and a `<collision>` tag (physical shape for interactions).
- **Add Sensors and Plugins:** Incorporate sensors (e.g., cameras, IMUs) and plugins to control your model's behavior.

Here is a very basic example of an SDF file for a simple cart:

XML

None

```
<?xml version='1.0'?>

<sdf version='1.7'>

  <model name='simple_cart'>

    <link name='chassis'>
```

```
<pose>0 0 0.1 0 0 0</pose>

<inertial>

  <mass>5.0</mass>

  <inertia>

    <ixx>0.04</ixx>

    <iyy>0.2</iyy>

    <izz>0.2</izz>

  </inertia>

</inertial>

<collision name='collision'>

  <geometry>

    <box>

      <size>0.4 0.3 0.1</size>

    </box>

  </geometry>

</collision>

<visual name='visual'>

  <geometry>

    <box>

      <size>0.4 0.3 0.1</size>

    </box>
```

```
        </geometry>

    </visual>

</link>

</model>

</sdf>
```

Key Points for the Template:

- **<pose>**: This essential tag defines the offset from the link's origin to its center of mass. Input the **Center of Mass** coordinates directly from your CAD software.
- **<uri>**: This specifies the file path to your mesh. If the mesh file is in the same folder as the .sdf file, you can use just its name.
- **<visual> vs. <collision>**: For simple parts, the same mesh can be used for both. For complex parts, use a high-detail mesh for the visual representation and a simpler, performance-friendly mesh for collision geometry.

For more details on SDF, refer to: <http://sdformat.org/spec?ver=1.12&elem=sdf>

Section 2: Simplify your CAD model

Follow these steps to optimize your CAD model, ensuring it's clean, efficient, and ready for use in real-time applications like simulators or renderers.

Step 1: Assign and Consolidate Materials 🧐

Before you export, it's crucial to organize your model's materials. The goal is efficiency, not perfect physical accuracy.

- **Assign Materials**: Make sure every part of your model has a material assigned to it.
- **Consolidate Materials**: This is the most important part. All parts that look the same should share the **exact same material**. For instance:
 - If you have ten bare aluminum parts, they should all use a single "Bare Aluminum" material.